

Triple Diffie-Hellman Key Exchange (3DH)



Ik_A

$ECDH_1$

$ECDH_2$

Ik_B



Ek_A

$ECDH_3$

SPk_B

$ECDH(Ik_A, SPk_B)$

$ECDH(Ek_A, SPk_B)$

$k_{root} = HKDF(ECDH_1, ECDH_2, ECDH_2)$

$ECDH(Ek_A, Ik_B)$

Double Ratchet Protocol

- The root key k_{root} **does not encrypt messages**
- Use **for deriving a series of keys** $k_1, k_2, k_3 \dots$ that will be used to encrypt (send) and decrypt (receive) each message m_1, m_2, m_3